

SET:

Küme birimdir

$S \rightarrow$ Set

$$S = \{0, 1, 2, 3, \dots\}$$
$$\{a, b, \dots, z\}$$

$$S = \{i, i > 0, i \text{ çift}\}$$

$\cup$, $\cap$, $\bar{}$
 $\downarrow$ $\downarrow$ $\downarrow$
Birleşim Kesişim fark

$$S_1 \cup S_2 = \{x : x \in S_1 \text{ or } x \in S_2\}$$

$$S_1 \cap S_2 = \{x : x \in S_1 \text{ and } x \in S_2\}$$

$$S_1 - S_2 = \{x : x \in S_1 \text{ and } x \notin S_2\}$$

$$\bar{S} = \{x : x \in U_1 \text{ and } x \notin S\}$$

$$\{a, b, c\} \rightarrow S$$

$$S \cup \emptyset = S - \emptyset = S$$

$$2^S = \{\emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{b, c\}\}$$

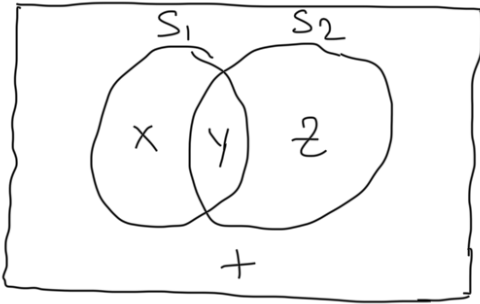
$$S = S_1 \times S_2 = \{(x, y) : x \in S_1, y \in S_2\}$$

$$S_1 = \{2, 4\}$$

$$S_2 = \{5, 6, 7\}$$

$$S_1 \times S_2 = \{(2, 5), (2, 6), (2, 7), (4, 5), (4, 6), (4, 7)\}$$

? Örnek $S_1 \cup S_2 = \overline{S_1} \cap \overline{S_2}$
olduğunu gösteriniz



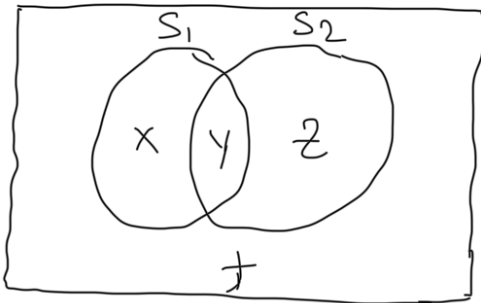
$$S_1 \cup S_2 = \{x, y, z\}$$

$$\{z, y\} \cup \{x, +\} = \overline{S_1} \cap \overline{S_2}$$

$$\Downarrow$$

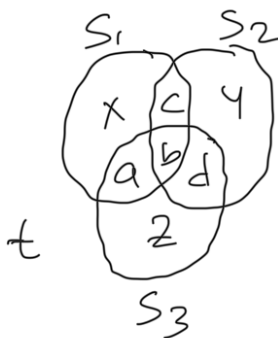
$$\{x, y, z, +\}$$

? Örnek $(S_1 \cap \overline{S_2}) \cup (\overline{S_1} \cap S_2) = \emptyset$ ispatı gösterin



$$\{x\} \cap \{z\} = \emptyset$$

Örnek $S_1 \times (S_2 \cup S_3) = (S_1 \cap S_2) \cup (S_1 \cap S_3)$
ispatını gösteriniz



$$S_1 = \{x, c, a, b\}$$

$$S_2 \cup S_3 = \{c, y, a, b, d, z\}$$

$$S_1 \cap S_2 = \{c, b\}$$

$$S_1 \cap S_3 = \{a, b\}$$

$$S = \{a, b, c\}$$

Örnek $\sqrt{2}$ 'nin irrasyonel olduğunu ispatlayınız.

$$4i-1$$

$\pi \rightarrow$ irrasyonel

$$\sqrt{2} = \frac{a}{b}$$

$$2b^2 = \underbrace{a^2}_{\rightarrow 2k}$$

$$2b^2 = 4k^2$$

$$\boxed{b^2 = 2k^2}$$

Örnek $\sqrt{3}$ 'ün irrasyonel olduğunu ispatlayınız

$$\sqrt{3} = \frac{a}{b}$$

$$3b^2 = \underbrace{a^2}_{\rightarrow 3k}$$

$$3b^2 = 9k^2$$

$$b^2 = 3k^2$$

Örnek p asal sayı ise $\sqrt{p}$ 'nin irrasyonel olduğunu ispatlayınız

$$\sqrt{p} = \frac{a}{b}$$

$$pb^2 = a^2$$

$$a = pk$$

$$pb^2 = p^2k^2$$

$$b^2 = pk^2$$

$$b = p_1$$

$$\sqrt{11}, \sqrt{5}, \sqrt{17}$$

Örnek $\sqrt{2} + \sqrt{3}$ 'ün irrasyonel olduğunu gösteriniz

$$\sqrt{2} + \sqrt{3} = \frac{a}{b}$$

$$2 + 2\sqrt{6} + 3 = \frac{a^2}{b^2}$$

$$5 + 2\sqrt{6}$$

$$2\sqrt{6} = \frac{a^2}{b^2} - 5$$

$$2\sqrt{6} = \frac{a^2 - 5b^2}{b^2}$$

$$\underbrace{\sqrt{6}}_{\downarrow \text{irrasyoneldir}} = \frac{a^2 - 5b^2}{2b^2}$$

DİLLER

Kelime gruplarından oluşan bir grameri takip eden sistemlerdir.

$$\Sigma \rightarrow \text{Alfabe}$$

$$\Sigma = \{a, b\}$$

$$W = ab, ba, aab, bbb$$

$u, v, v \rightarrow$ kelimeleri temsil eder

$$u = a_1 a_2 \dots a_n$$

$$v = b_1 b_2 \dots b_n$$

$$uv = a_1 a_2 \dots a_n b_1 b_2 \dots b_n$$

$$w^2 = a_n \dots a_2 a_1$$

$$|\lambda| = 0$$

$$\boxed{\lambda w = w \lambda = w}$$

$$w^0 = \lambda$$

$$|uv| = |u| + |v|$$

$$|a| = 1$$

$$|wa| = |w| + 1$$

$$\Sigma^* = \{\lambda, a, b, ab, ba, \dots\}$$

$$|\Sigma^*| = \infty$$

$$\Sigma^+ = \Sigma^* - \{\lambda\}$$

$$\Sigma = \{a, b\}$$

$$L = \{a^n b^n : n \geq 0\}$$

$$W = \{ \lambda, ab, aabb, aaabbb, \dots \}$$

$$\{ a, aa, aab \}$$

$$L = \{ a^n b^m : n \geq 1, m = 0 \text{ or } m = 1 \}$$

$$\bar{L} = S^* - L$$

$$L^R = \{ w^R : w \in L \}$$

$$\Sigma = \{ a, b \}$$

$$L = \{ ab, bb, aab \}$$

$$L^R = \{ ba, bb, baa \}$$

$$L = \{ a^n b^n : n \geq 1 \}$$

$$L^R = \{ b^n a^n : n \geq 1 \}$$

$$L^0 = \{x\}$$

$$L^1 = L$$

$$L^2 = L \cup L$$

$$L^3 = L \cup L \cup L$$

$$L^* = \{x\} \cup L^1 \cup L^2 \dots$$

$$L^+ = L^1 \cup L^2 \cup \dots$$

$$L = \{a^n b^m \mid n \geq 1, m \geq 1\}$$

$$L^e = \{b^m a^n \mid n \geq 1, m \geq 1\}$$

GRAMER

Bir dilin mekanizmasıdır.

Belirlenen iskelete Atıf yapar.

$$G = (V, T, S, P) \rightarrow \text{tuple}$$

Variable, Terminal sembolü, Boşlangıç değişkeni, üretim kuralları

$$X \rightarrow y$$

$$(UVT)^*$$

$$(UVT)^+$$

$$W = U \times V$$

$$Z = U \cup V$$

$$W \Rightarrow Z$$

$$W_1 \Rightarrow W_2 \Rightarrow W_n$$

$$\boxed{V_1 \Rightarrow V_n}^*$$

$$L(G) = \{u \in T^* \mid S \Rightarrow^* u\}$$

$$G = (\Sigma^4, d, a, b, c, p)$$

$$P \begin{cases} S \rightarrow a \, b \\ S \rightarrow \lambda \end{cases}$$

$$S \rightarrow a$$

$$L = \{a^n b^n \mid n \geq 0\}$$

$$\textcircled{1} S \rightarrow \lambda \rightarrow w \in \{\lambda\}$$

$$\textcircled{2} S \rightarrow aSb \quad w \in \{ab\}$$

$$\rightarrow a a S b b \quad w = \{aabb\}$$

$$a a a S b b b \quad w = \{aaabbb\}$$

$$L = \{a^n b^{n+1} \mid n \geq 0\}$$

gramerini bulursak λ üretebilir mi?

w

b
 abb

$aabb$

$aaabbb$

$$\begin{array}{l} S \rightarrow b \quad \mid \quad S \rightarrow b \mid aSb \quad \mid \quad S \rightarrow Ab \\ S \rightarrow aSb \quad \mid \quad A \rightarrow \lambda \mid aAb \end{array}$$

~~Örnek~~

$$\Sigma = \{a, b\} \quad L = \{w \mid n_a(w) = n_b(w)\}$$

Dili için gramer bulunur.

$$S \rightarrow \lambda$$

$$S \rightarrow aSb \mid bSa \mid abS \mid baS \mid Sba \mid Sab$$

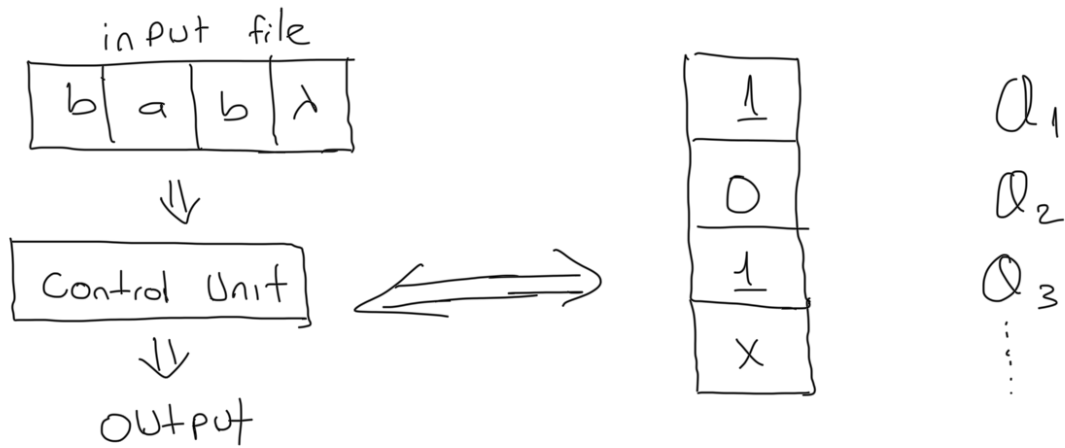
x $aaabaaa$

x $aaabbaa$

$$S \rightarrow aSb$$

$$\rightarrow aaabSb$$

Otomat



$Q_1 \xrightarrow{f} Q_2$ (Next State)
 $f \rightarrow$ transition function

Örnek

$w = aabbaab$ iken ww^Rw içinde kaç adet qab vardır?

$$w^R = baabbaa$$

aabbabbaabbaaaabbab
 1 1

2 adet vardır.

Örnek

$(w^R)^R = w$ teorisini $w \in \Sigma^*$ için ispatlayınız

$$\Sigma = \{0, 1\}$$

$$w = \{0, 1\} \rightarrow 01, 101, 1110$$

$$(110)^R = w$$

$$011 = 011$$

Örnek $\Sigma = \{a, b\}$ $L = \{aa, bb\}$
için L' 'yi tanımlayınız.

$$L = \{a^n \text{ or } b^n, n = 2\}$$

$$L = \{w \in \Sigma^*, |w| = 2\}$$

ab
ba

Örnek $L = \{a^n, n = \text{çift}\}$
için gramer yazınız

$$S \rightarrow \lambda$$

$$S \rightarrow Saa$$

Örnek $\Sigma = \{a, b\}$ için tüm kelimelerde
2 adet "a" içeren grameri bulunuz

Kelimeler

aa

aba

baa

abba

$$S \rightarrow aa \mid AaAaAa$$

$$A \rightarrow bA \mid \lambda \mid b$$

⇒ en az 2 adet 'a' içeren gramer yazın

$S \rightarrow AaAaA$
 $\rightarrow bAaAa bA$
 $\rightarrow bbAabAa b bA$
 $\rightarrow b b a b a b b$

$S \rightarrow AaAaA$
 $A \rightarrow Ab / Aa / \lambda$

⇒ 3'ten fazla 'a' olmayan kelimelerin gramerini yazınız.

Kelimeler
 λ

$bab, \dots$

$a \dots b \dots a \dots b$

$bb \dots a a a b b b \dots$

$S \rightarrow \lambda / AaA / BaBaB$
 $A \rightarrow bA / \lambda / a / CaCaCaC$
 $B \rightarrow bB / \lambda / a$
 $C \rightarrow bC / \lambda$

$S \rightarrow \lambda / AaA / AaAaAa /$

$|AaAaAaA|$

$A \rightarrow \lambda / Ab /$

~~Örnek~~

$S = \{a^i b^j\}$ için tüm kelimelerde a ile başlayan b ile biten kelimelerin gramerini yazınız

$S \rightarrow aAb$

$A \rightarrow aA / bA / \lambda$

⇒ Tüm kelimelerde çift sayıda b içeren gramer nedir?

$$S \rightarrow \lambda \mid AbAb \mid$$

$$A \rightarrow aA \mid \lambda \mid bbaA$$

Örnek

$$S \rightarrow aaA$$

$$A \rightarrow bS$$

$S \rightarrow \lambda$ için dil yazınız.

$$\lambda \quad aaA$$

$$aab \quad aabS$$

$$aqbab \quad aabaaA$$

$$L = \{ (aab)^n : n \geq 0 \}$$

Örnek

$$S \rightarrow Aa \quad G = (\{S, A, B\})$$

$$A \rightarrow B \quad \{a\}, S, P$$

$$A \rightarrow Aa \quad \text{ için dil yazınız}$$

$$S \rightarrow Aa$$

$$\rightarrow Ba$$

$$Aa$$

$$Aaa$$

⋮

Sonsuza gide o yüzden

tanımsızdır

Örnek $L = \{ a^n b^{n-2} \mid n \geq 3 \}$ için gramer yazınız

Kelimeler

aaa b

aaaa bb

aaaaa bbb

$S \rightarrow aAb$

$A \rightarrow aa \mid aAb$

Örnek $S \rightarrow aSb \mid ab \mid \lambda$

$S \rightarrow aaSbb \mid aSb \mid ab \mid \lambda$

iki gramerin eşitliğini gösteriniz.

$S \rightarrow aSb \mid ab \mid \lambda \rightarrow \lambda, a, b$

$S \rightarrow aaSbb \mid aSb \mid ab \mid \lambda \rightarrow \lambda, a, b$

$L = \{ a^n b^n \mid n \geq 0 \}$

Örnek $\Sigma = \{ a, b \}$
 $S \rightarrow aSb \mid bSa \mid a$

için biçimsel dili yazınız

Kelimeler

$\left. \begin{array}{l} a \\ aab \\ aasbb \end{array} \right\} \begin{array}{l} a^{n+1} b^n \\ n \geq 0 \end{array}$

$\left. \begin{array}{l} bsa \\ bbsaa \\ bbbbaaa \end{array} \right\} \begin{array}{l} b^n a^{n+1} \\ n \geq 0 \end{array}$

$L = \{ a^{n+1} b^n a^n b^n a^{n+1} \mid n \geq 0 \}$

Deterministic finite ACCEPTER

$M \rightarrow$ Otomat

Sonlu sayıda durum vardır

Bir durumdan diğerine geçiş, girişe bağlıdır.

Çıktı üretir

$M(Q, \Sigma, \delta, q_0, f)$

$Q \rightarrow$ Sonlu durumlar

$\Sigma \rightarrow$ Alfabe (giriş)

$\delta \rightarrow$ Geçiş fonksiyonu

$q_0 \rightarrow$ Başlangıç Durumu

$f \rightarrow$ Sonlu durumlar

$\delta \rightarrow Q \times \Sigma \rightarrow Q$

$q_0 \in Q$

$f \in Q$

$f \subseteq Q$

q_0
 $\downarrow$
1 ade

f
 $\downarrow$
1 den fazla olabilir.

$\delta(q_0, a) = a^n$

$M = (\{q_0, q_1, q_2\}, \{0, 1\}, \delta, q_0, \{q_1\})$

$$f(q_0, 0) = q_0$$

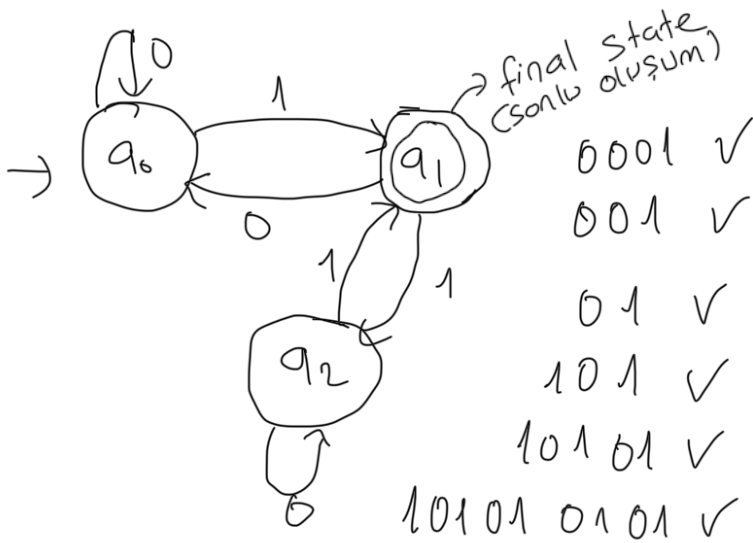
$$f(q_0, 1) = q_1$$

$$f(q_1, 0) = q_0$$

$$f(q_1, 1) = q_2$$

$$f(q_2, 0) = q_2$$

$$f(q_2, 1) = q_1$$



0001 ✓

001 ✓

01 ✓

101 ✓

10101 ✓

101010101 ✓

11 X

0 X

000 X

00 X

1111 X

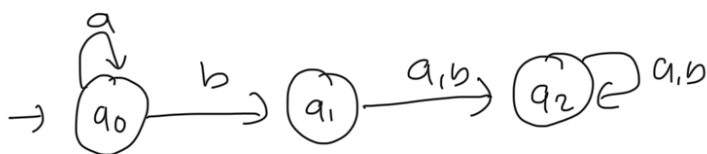
$$f^*(q_0, ab) = q_2$$

$$f^*(Q \times \Sigma^*) = Q$$

$$f^*(q, wa) = f(f^*(q, w), a)$$

$$L(M) = \{w \in \Sigma^* \mid \delta^*(q_0, w) \in f\}$$

$$L = \{a^n b \mid n \geq 0\}$$



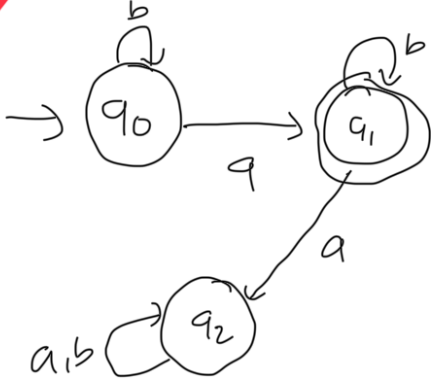
(Trap State)

Tuzak durumu

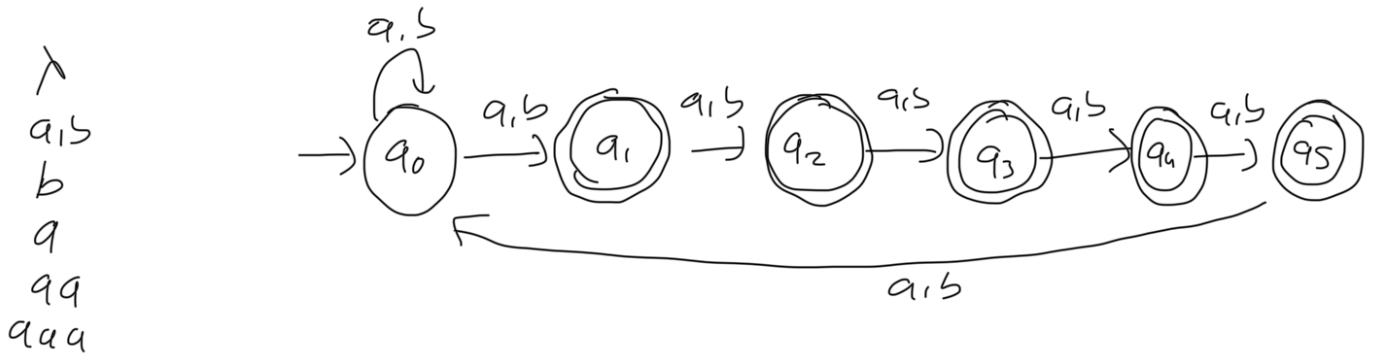
$$\begin{aligned} f(q_0, a) &= q_0 \\ f(q_0, b) &= q_1 \\ f(q_1, a) &= q_2 \end{aligned}$$

$$\begin{aligned} f(q_1, b) &= q_2 \\ f(q_2, a) &= q_2 \\ f(q_2, b) &= q_2 \end{aligned}$$

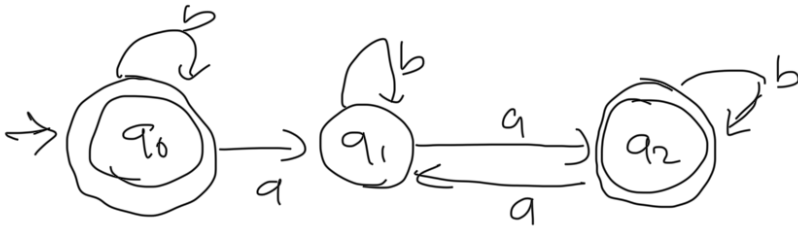
✓ Örnek $\Sigma = \{a, b\}$ alfabelerinden oluşan DFA'da tek 'a' olsun



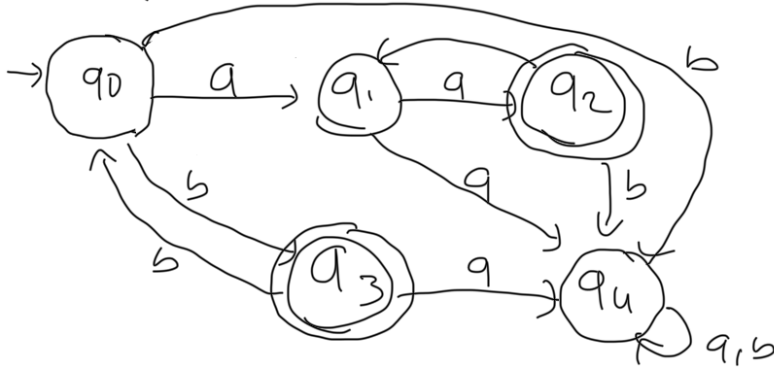
✓ Örnek $\Sigma = \{a, b\}$ için $|w| \leq 5$ 'e uyandıran DFA çiziniz.



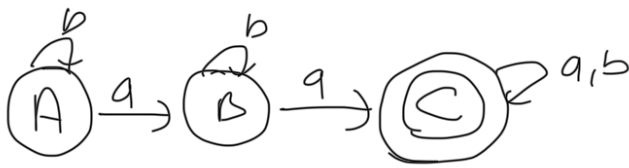
Örnek $\Sigma = \{a, b\}$ 'den üretilen kelimelerde çift sayıda "a" olsun DFA çiziniz



Örnek $\Sigma = \{a, b\}$ üretilen w için $|w| \bmod 4 = 0$ (çift)
 $|w| \bmod 4 = 1$ için DFA çiziniz.

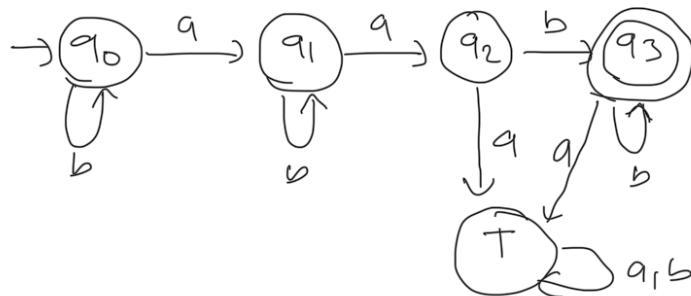


Örnek $\Sigma = \{a, b\}$ en az 2 adet 'a' içeren $|w|$ içeren
 $|w|$ için DFA çiziniz



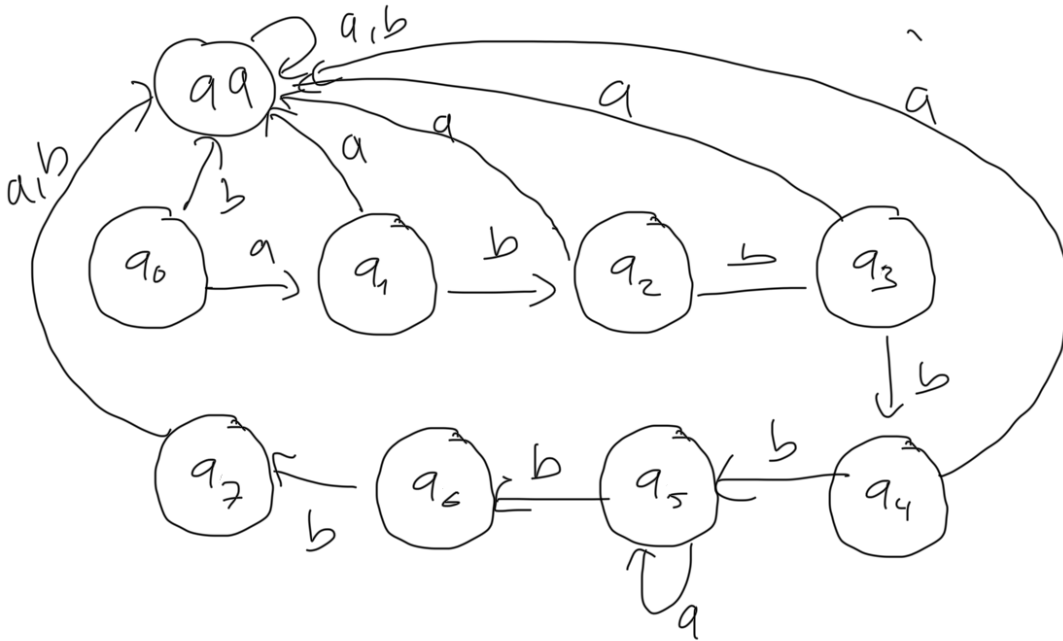
Örnek $\Sigma = \{a, b\}$ üretilen kelimelerden en az
 1 adet 'b' ve 2 adet 'a' içeren DFA yazınız

Kelimeler
 b a a
 a a b b
 a b a b b b
 b a a b b b ...



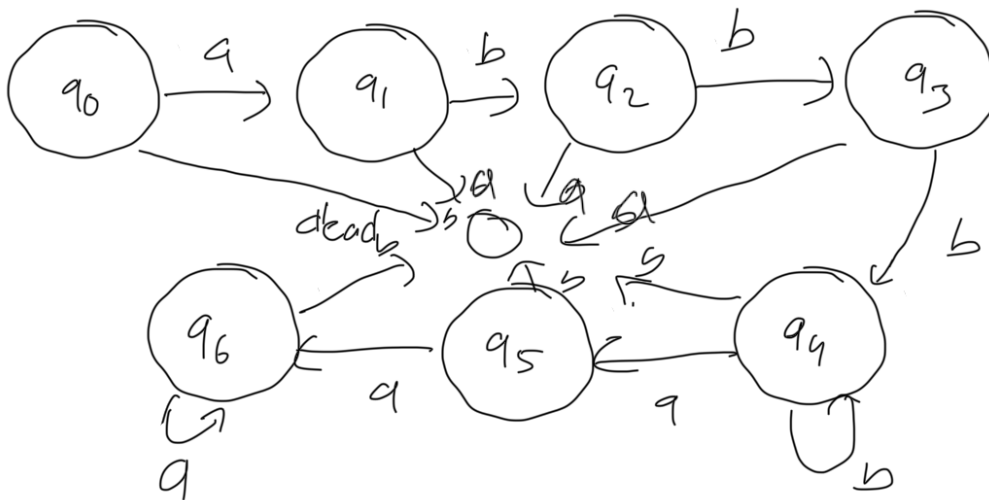
Örnek

$\Sigma a b^4 w b^2 : w \in \Sigma a, b \Sigma^*$
in dfa'yı çiziniz



Örnek

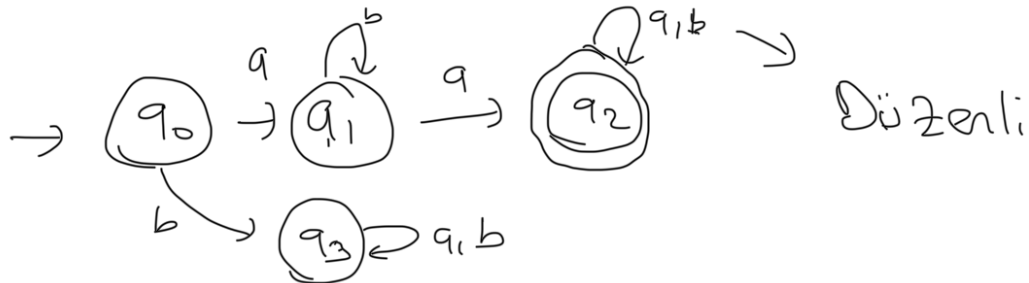
$L = \{ a, b^n, a^m \mid n \geq 3, m \geq 2 \}$
İçin DFA çiziniz



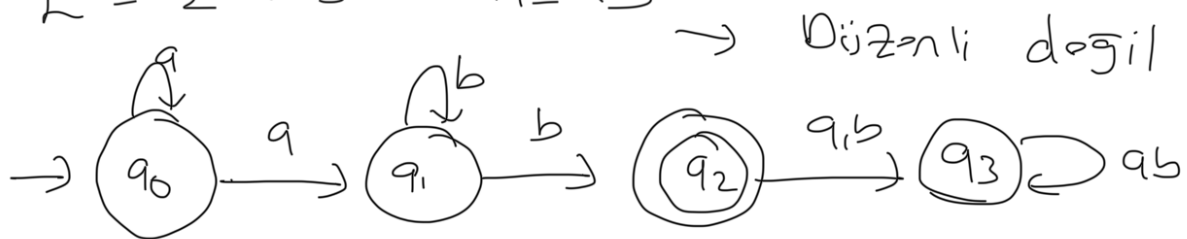
Regular Languages

$L = L(M) \rightarrow$ deterministik ise dil düzenlidir

$$L = \{ a^i b^j : i, j \geq 1 \}$$



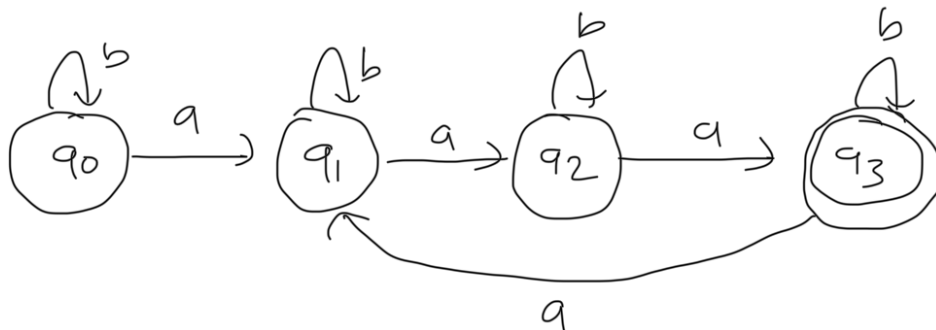
$$L = \{ a^n b^n : n \geq 1 \}$$



örnek

$$\Sigma = \{ a, b \}$$

$$L = \{ w : n_a(w) \bmod 3 < 1 \}$$
 DFA Çiniz



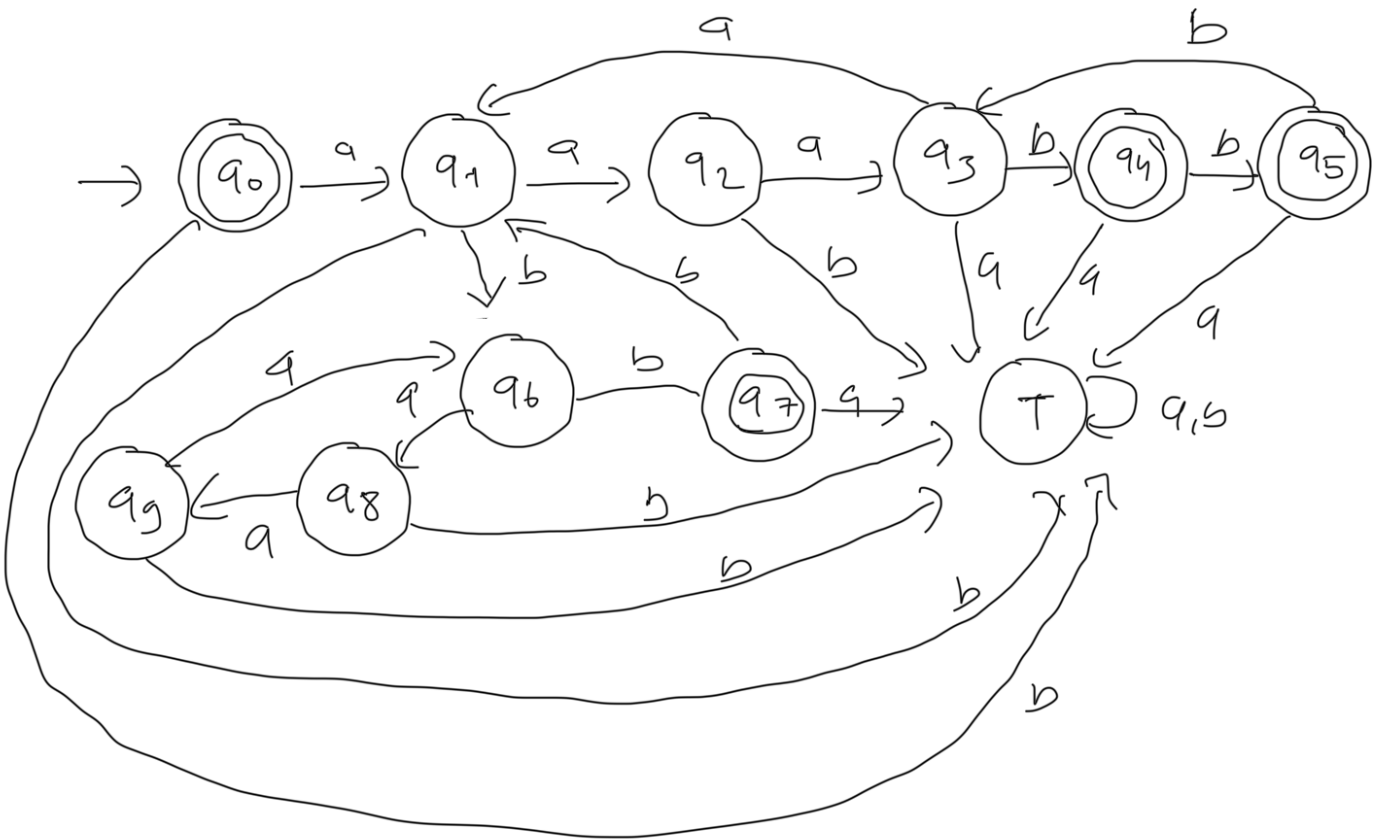
Sınavda
gelebilir
örnek !

$$\Sigma = \{a, b\}$$

$$L = \{w \mid n_a(w) \bmod 3 < n_b(w) \bmod 3\}$$

için DFA çiziniz

n_a	n_b
0	1, 2
1	2

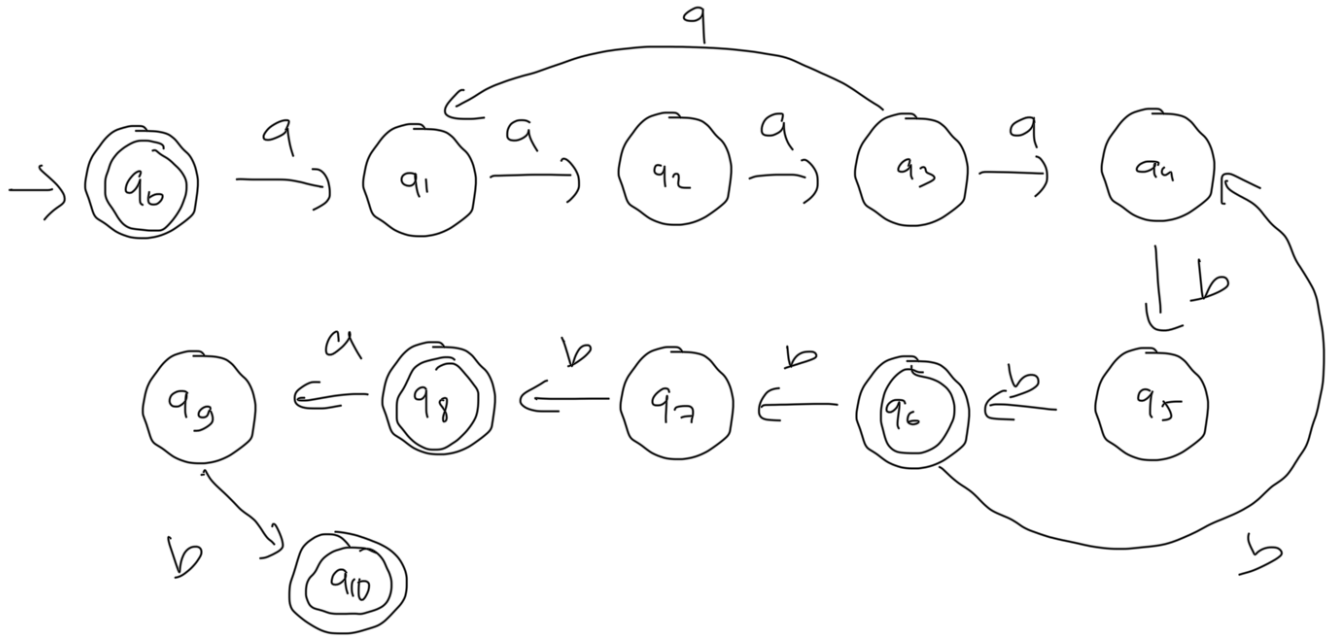


örnek

$$\Sigma = \{a, b\}$$

$$L = \{w \mid (n_a(w) - n_b(w)) \bmod 3 = 0\}$$

n_a	n_b
$3k$	$3k$
$3k+1$	$3k+1$
$3k+2$	$3k+2$



Örnek

$$\Sigma = \{a, b\}$$

$$L = \{w : (n_a(w) + 2 n_b(w)) \bmod 3 = 0\}$$

DFA çiziniz.

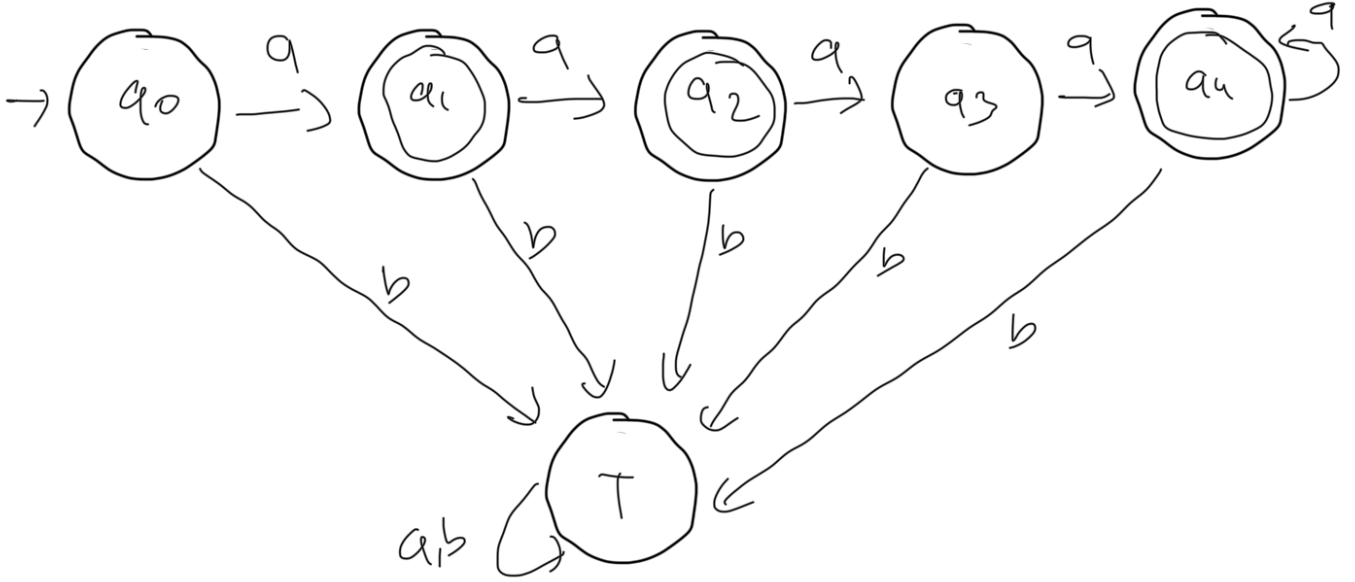
<u>n_a</u>	<u>n_b</u>		<u>n_a</u>	<u>n_b</u>
$3k+1 \left\{ \begin{array}{l} 1 \\ 1 \\ 1 \\ 1 \end{array} \right.$	1	$\left. \begin{array}{l} 1 \\ 4 \\ 7 \\ 10 \end{array} \right\} 3k+1$	2	1
	4		5	2
	7		0	6
	10			

Çözümü bir önceki soru ile aynı,
aynı soruyu farklı bir biçimde sormuş

Örnek

$$L = \{ a^n \mid n \geq 0, n \neq 3 \}$$

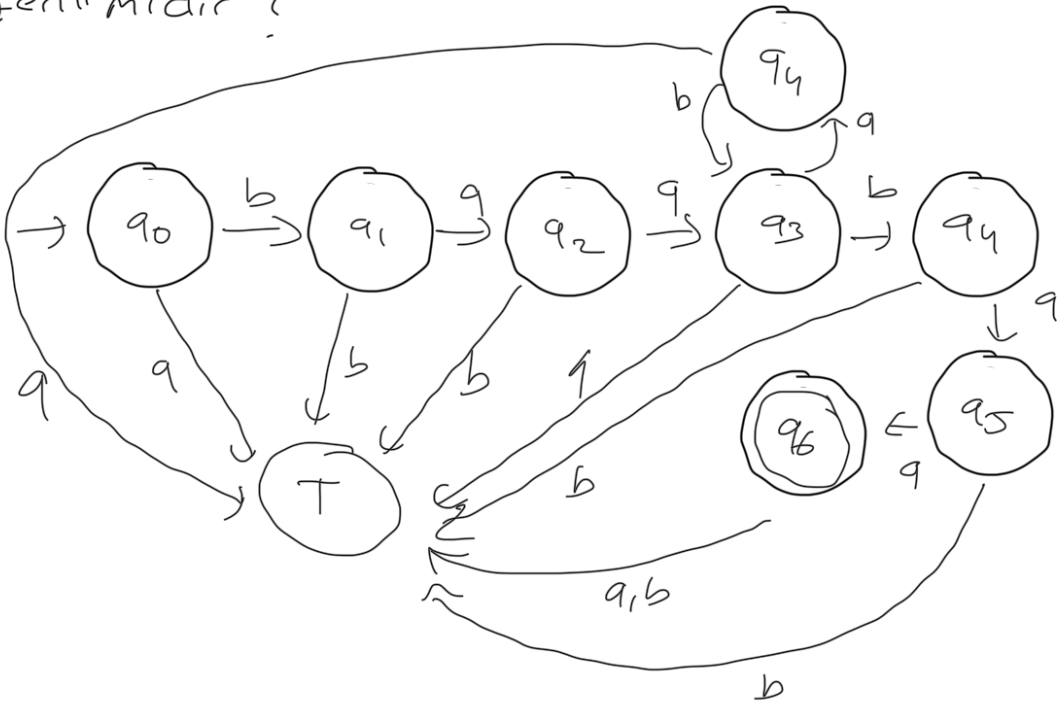
için düzenli olduğunu gösteriniz.



Örnek

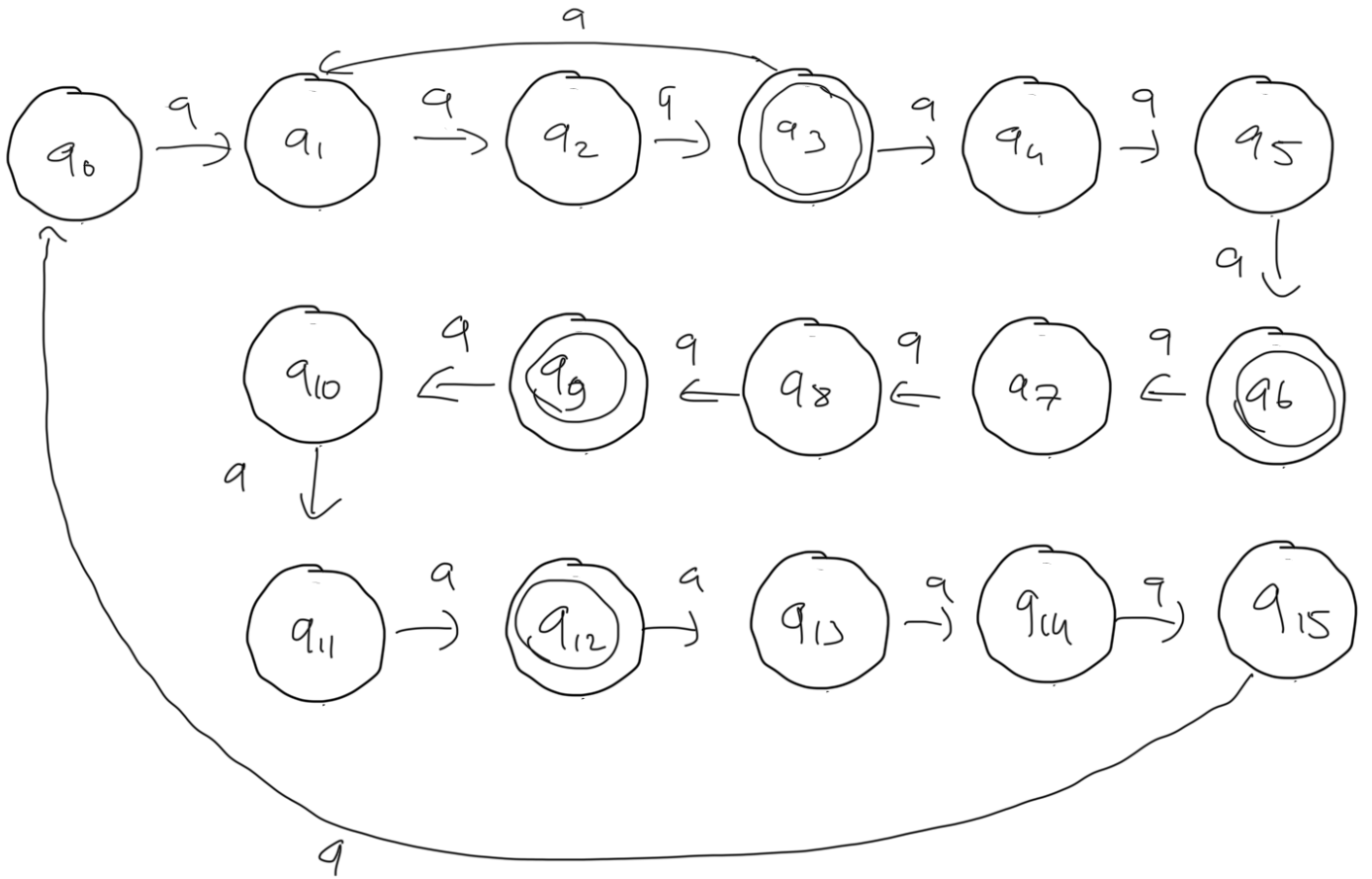
$$L = \{ vwv : v, w \in \{a, b\}^*, |v| = 3 \}$$

Düzenli midir ?



Örnek

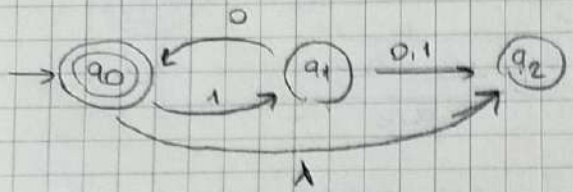
$L = \{ a^n : n = 3k, n \neq 5k \}$
için Düzenli denir mi?



Soru: Bir NFA tasarlayın, bu NFA tüm doğal sayıları kabul etsin.

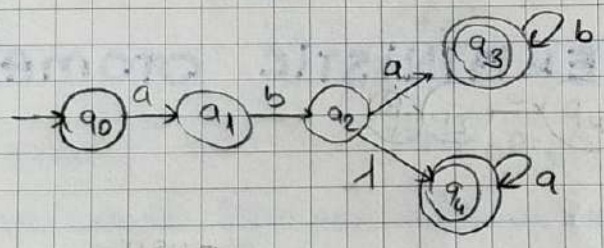


Soru:



$\delta^*(q_0, 1011) = q_2$
 $\delta^*(q_1, 01) = q_1$
 $\delta^*(q_0, 1010) = q_0, q_2$
 $\delta^*(q_1, 00) = \text{herhangi bir yere gitmez.}$

Soru: $S = \{abab^n : n \geq 0\} \cup \{aba^n : n \geq 0\}$ en fazla 5 durumlu NFA çizilmiştir.

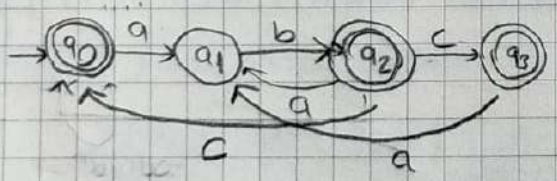


Soru:

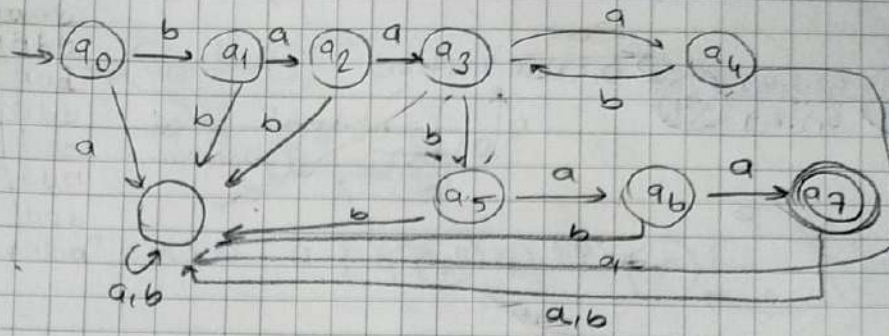
$L = \{ab, abc\}^*$ buna ait NFA'yı nasıl çizeriz.
 → kombinasyon demek

Kelimeler

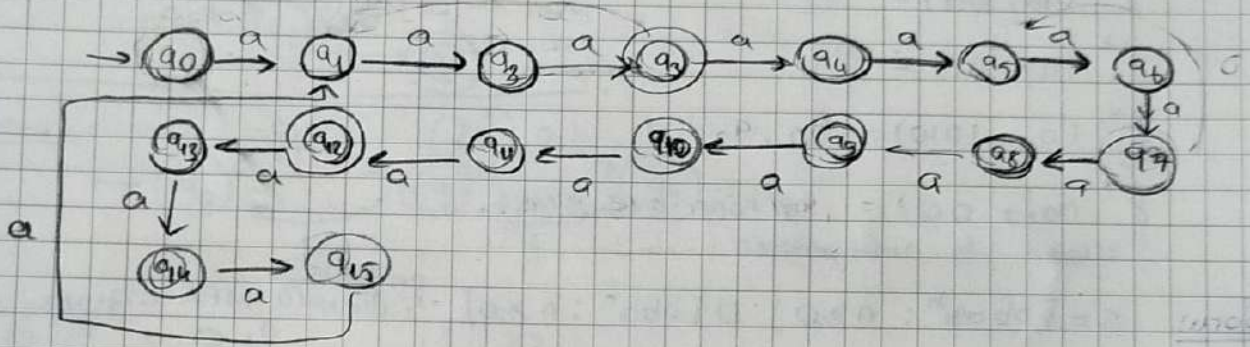
- ab
- λ
- ababc
- abcab
- abab
- ⋮



Soru: $L = \{vwv : v, w \in \{a,b\}^+, |v|=3\}$ düzenli midir?



Soru: $L = \{a^n : n \equiv 3k, n \neq 5k\}$ için düzenli denir mi?



NON-DETERMINISTIC OTOMAT

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q : Sonlu durumlar

Σ : Alfabe

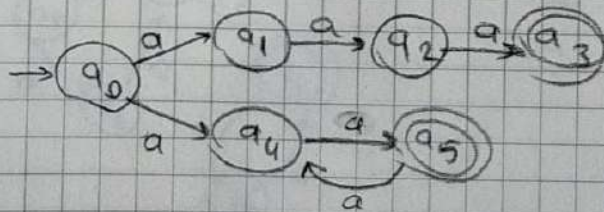
δ : Geçiş fonksiyonu

q_0 : başlangıç durumu

F : Sonlu durumlar.

$$\delta: Q \times (\Sigma \cup \{\epsilon\}) \rightarrow 2^Q$$

$$\delta(q_0, a) = \{q_1, q_2\}$$

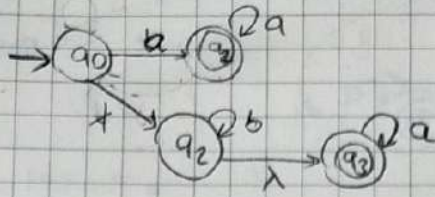


$$L = \{a^n : n = 3 \text{ or } n = 2K, K \geq 1\}$$

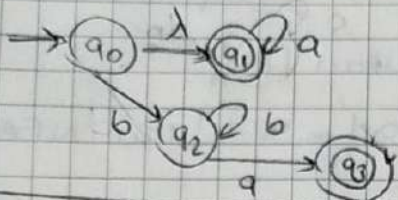
Not: tuzak durum yok!

Soru: $L = \{a^n : n \geq 1\} \cup \{b^m a^k : m \geq 0, k \geq 0\}$ için NFA çiziniz.

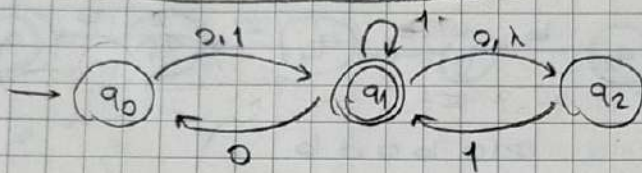
a
ba
bbba
b bbaaa



Soru: $L = \{a^n : n \geq 0\} \cup \{b^n a : n \geq 1\}$ için NFA çiziniz.



Soru:



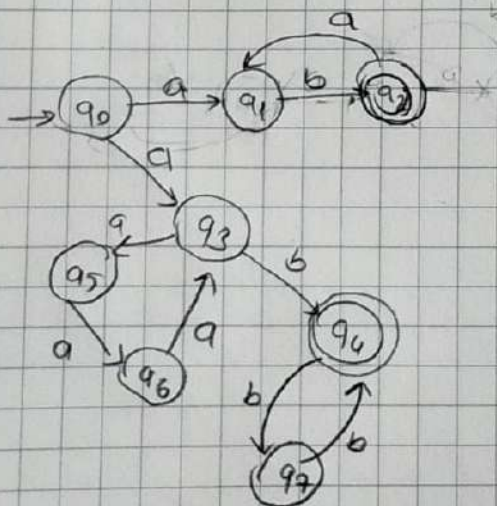
otomatının dili nedir?

kelimelere

- $1^n = 01^n$ $\{1^n : n \geq 1\}$
- $01 = 001$
- $000... = 0^n$ $100...$
- $11 = 01000$

$$L = \{w \in \Sigma^+, |w| \geq 1\}$$

Soru: $L = \{(ab)^n : n \geq 1\} \cup \{a^k b^m : k \bmod 3 = 1, m \bmod 2 = 1\}$ için NFA çiziniz.



Soru:

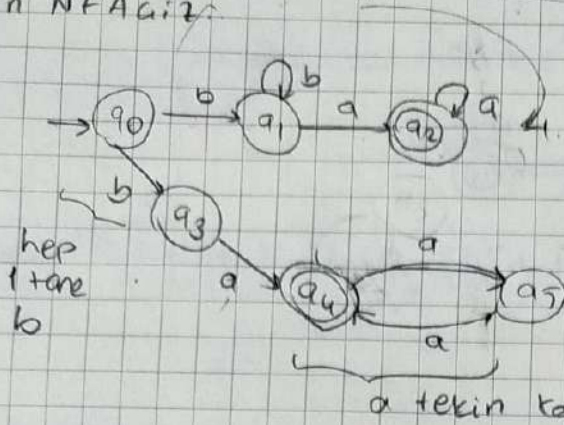
$$L = \{ b^n a^m : n, m \geq 1 \} \cup \{ b^k a^r : r, k \geq 1, n = k+r, r \bmod 2 = k \}$$

icin NFA çiz.

kelimeler

bba

bb



Sağlanamaz.	3	1
	5	1
	7	1

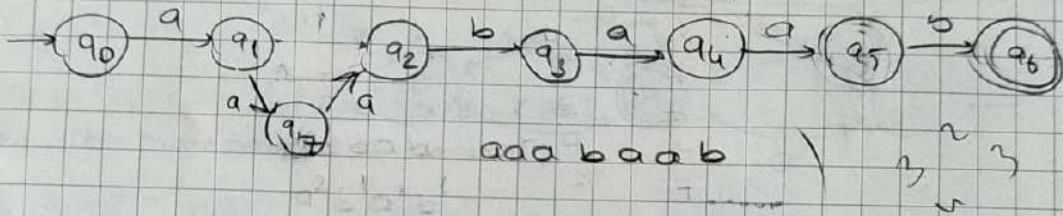
• NFA'de
durumu
burda
sağlanamaz.

Soru:

$$L = \{ a^n b^k a^m b^r : k \bmod r = 0, n \bmod m = 1 \}$$

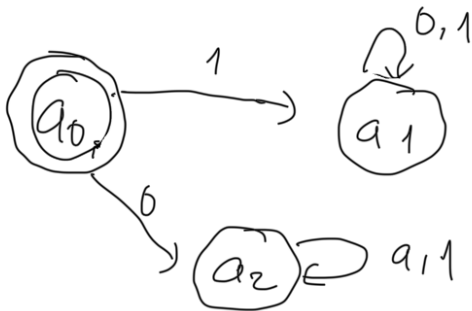
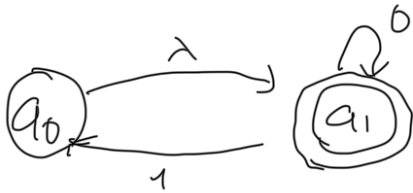
icin NFA çiz.

aa

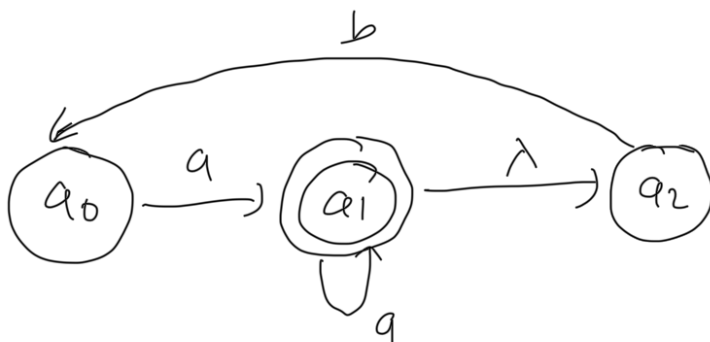


Deterministik ve Non deterministik Otomatların eşitlikleri

M_1 ve M_2 için,
 $L(M_1) = L(M_2)$
 ancak aynı dil kabul ediliyorsa mümkündür

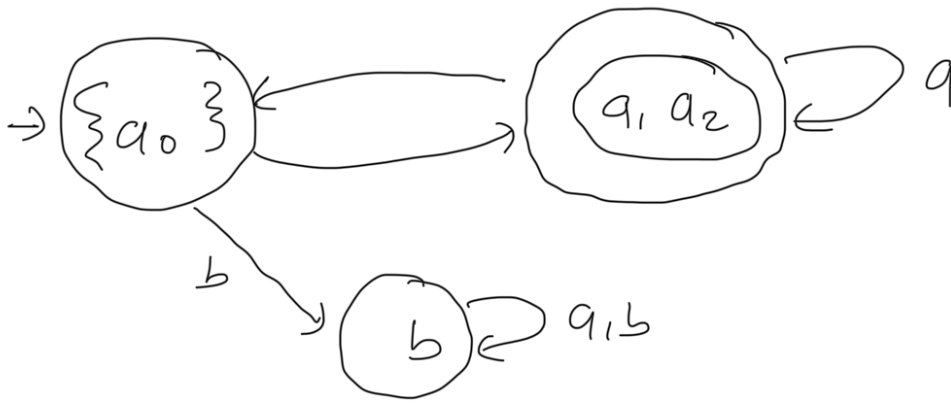


$$L(\{q_0\}, a) = \{q_1, q_2\}$$



$$\delta(\{q_1, q_2\}, a) = \{q_1, q_2\}$$

$$\delta(\{q_1, q_2\}, b) = \{q_0\}$$



L için $m_n (Q_n, \Sigma, \delta_n, q_0, f_n)$ için
 $m_D (f_n \Sigma, \delta_n, \{a_0\}, f_D)$
 $L = L(m_D)$

Dönüşüm Adımları

1. Q_0 çiz $\{a_0\}$ ile başla
 $\{a_1, a_2, \dots, a_k\}$

Birden fazla durum çıkabilir

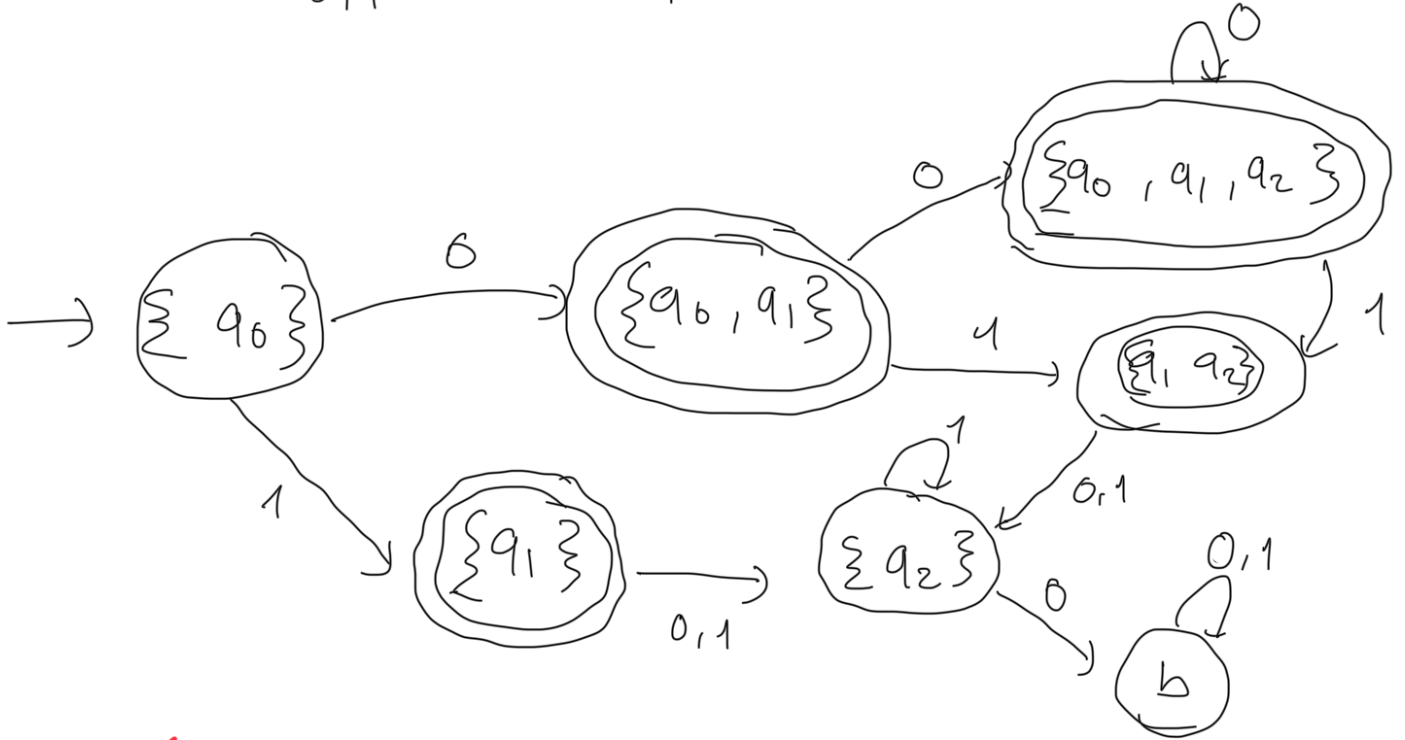
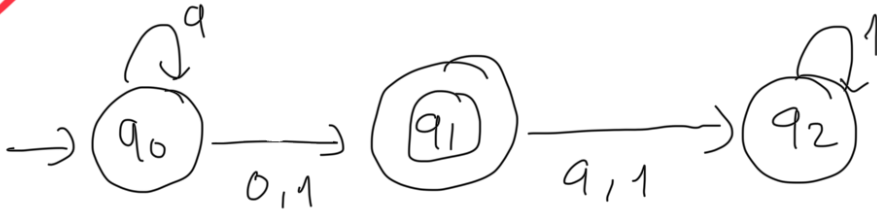
$\delta^* N(a_i, a)$
 $\delta^* N(a_j, a)$
 $\delta^* N(a_k, a)$

Q_0 için $af \in F_N$

$m_n \lambda$ kabul ediliyorsa

$\{a_0\} \neq$ elemanıdır

örnek NFA'yı DFA'ya dönüştür.



örnek

$$\delta(q_0, a) = \{q_0, q_1\}$$

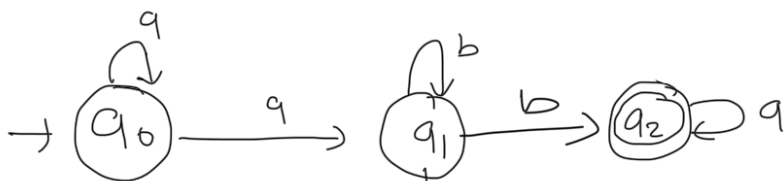
$$\delta(q_1, b) = \{q_1, q_2\}$$

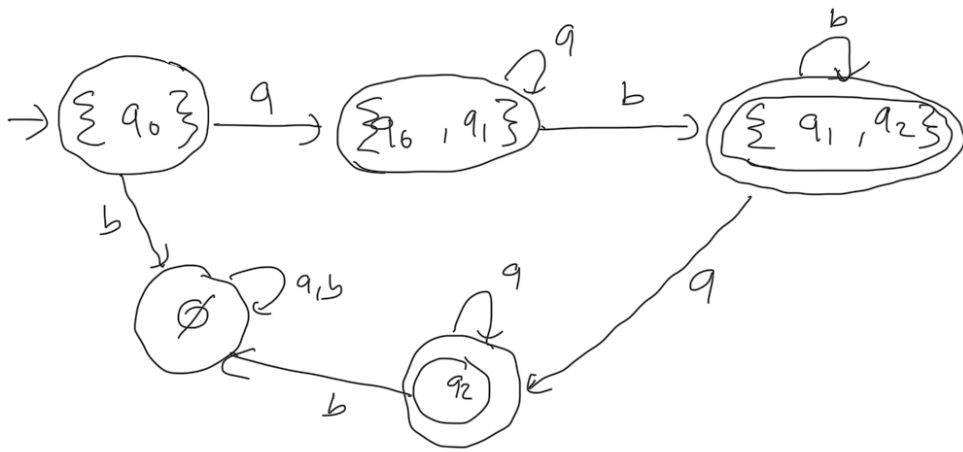
$$\delta(q_2, a) = \{q_2\}$$

$$q_0 = \{q_0\}$$

$$F = \{q_2\}$$

NFA'yı DFA'ya çeviriniz.





örnek

NFA'yı DFA'ya dönüştür

$$\mathcal{I}(q_0, a) = \{q_0, q_1\}$$

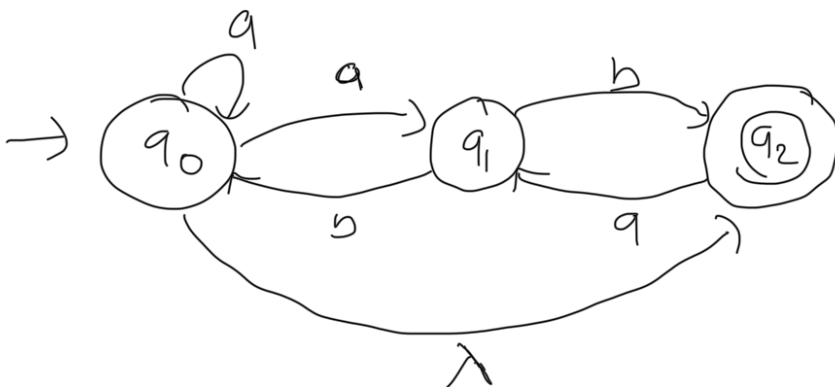
$$\mathcal{I}(q_1, b) = \{q_0, q_2\}$$

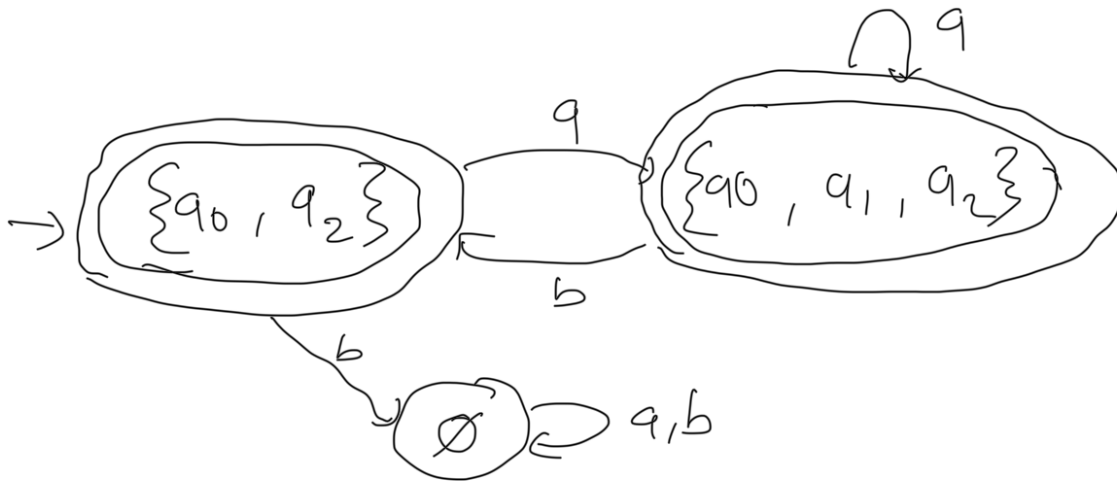
$$\mathcal{I}(q_2, a) = \{q_1\}$$

$$\mathcal{I}(q_0, \lambda) = \{q_2\}$$

$$q_0 = \{q_0\}$$

$$f = \{q_2\}$$





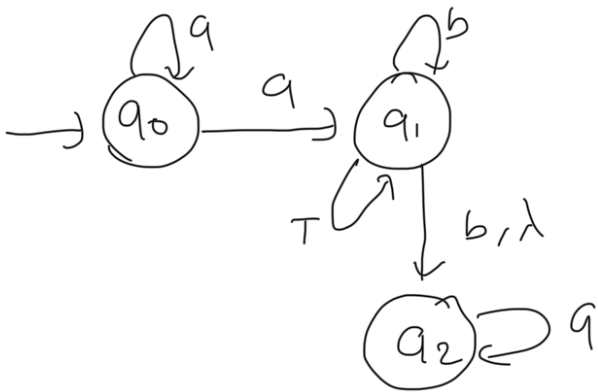
Örnek NFA'yı DFA'ya dönüştür.

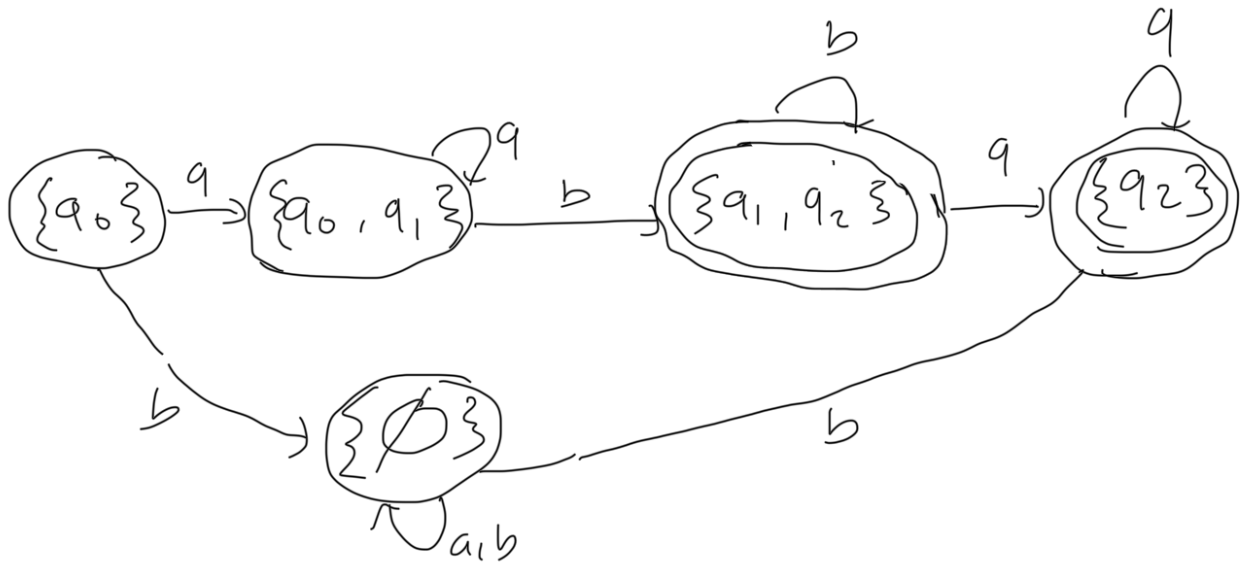
$$\delta(q_0, a) = \{q_0, q_1\}$$

$$\delta(q_1, b) = \{q_1, q_2\}$$

$$\delta(q_2, a) = \{q_2\}$$

$$\delta(q_1, \lambda) = \{q_1, q_2\}$$





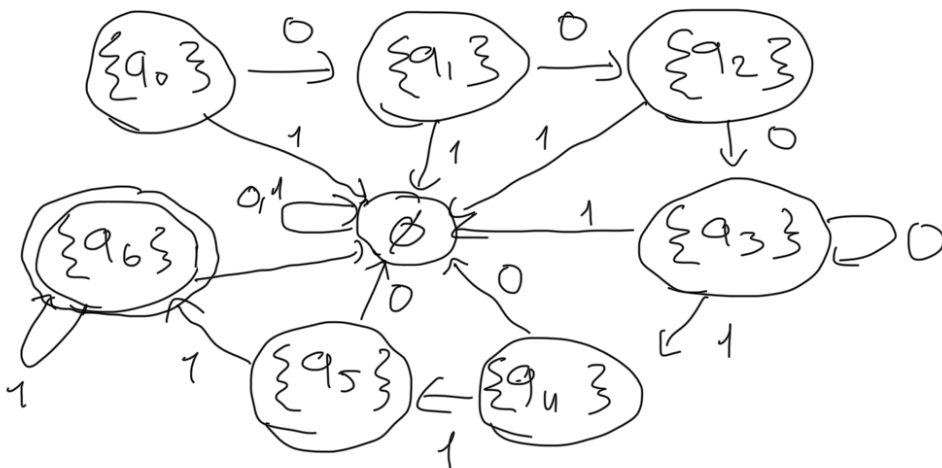
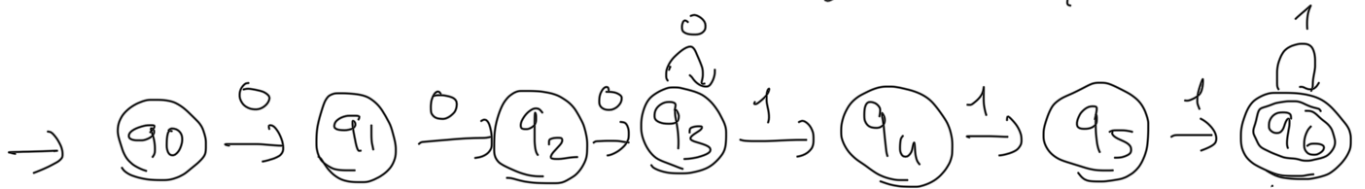
örnek

$$L = \{0^n 1^m\} \cap \{0^p 1^r\}$$

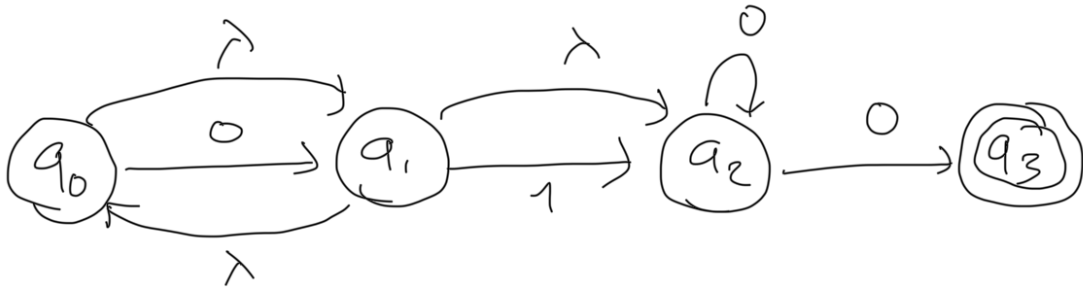
$$n, m \geq 2, p, r \geq 3$$

$$0^n 1^m \rightarrow n, m \geq 3$$

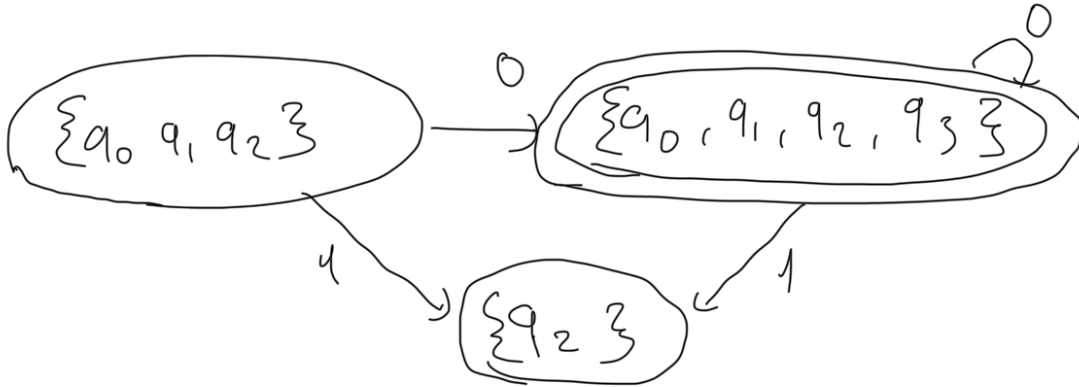
İçin çizilen NFA'yı DFA'ya dönüştürün



Örnek



NFA'yı DFA'ya dönüştürünüz.

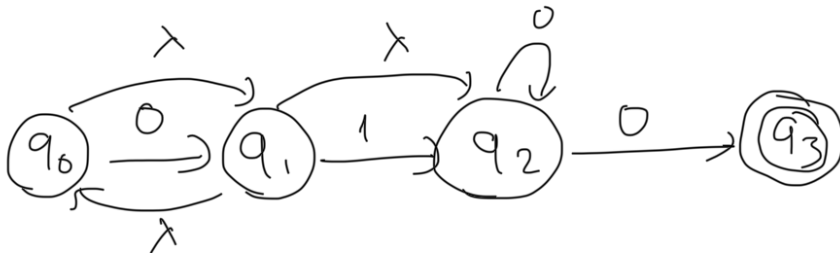


Durumların Azaltılması

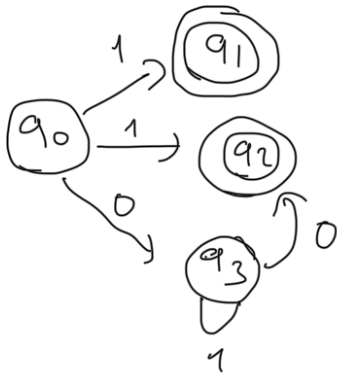
$\delta^*(p, w) \in F$, $\delta^*(q, w) \in f$ > p, q ayırt edilemez

$\delta^*(p, w) \notin f$, $\delta^*(q, w) \notin f$ > p, q ayırt edilemez

Örnek



NFA'yı DFA'ya dönüştürünüz.



Procedure

- ① Erişilmeyen durumları sil
- ② Tüm durum çiftlerini düşün (p, a)
- ③ $(p, a) \rightarrow a \in \Sigma$

$$\begin{aligned} \delta(p, a) &= pq > (pa, qa) \\ \delta(q, a) &= qa \end{aligned}$$